

Question	Marking Guidance	Mark	Comments
3.1	1. Co-transport; 2. Uses (hydrolysis of) ATP; 3. Sodium ion and proton bind to the protein; 4. Protein changes shape (to move sodium ion and/or proton across the membrane);	3 max	3. Accept 'Na ⁺ and H ⁺ bind to protein' but do not allow incorrect chemical symbols
3.2	1. Tenapanor/(Group) B /drug causes a <u>significant</u> increase; OR There is a <u>significant</u> difference with Tenapanor/drug/between A and B ; 2. There is a less than 0.05 <u>probability</u> that the difference is due to <u>chance</u> ; 3. (More salt in gut) reduces water potential in gut (contents); 4. (so) less water absorbed out of gut (contents) by <u>osmosis</u> OR Less water absorbed into cells by <u>osmosis</u> OR Water moves into the gut (contents) by <u>osmosis</u> . OR (so) water moves out of cells by <u>osmosis</u> ;	4	1. and 2. Reject references to 'results' being significant/due to chance once only. 2. Do not credit suggestion that probability is 0.05% or 5. 2. Accept 'There is a greater than 0.95/95% probability that any difference between observed and expected is not due to chance'
3.3	1. (Higher salt) results in <u>lower</u> water potential of tissue fluid; 2. (So) less <u>water</u> returns to capillary by osmosis (at venule end); OR 3. (Higher salt) results in <u>higher</u> blood pressure/volume; 4. (So) more fluid pushed/forced out (at arteriole end) of capillary;	2	For 'salt' accept 'sodium ions'. Do not allow mix and match of points from different alternative pairs 3. Accept higher hydrostatic pressure.
